Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 279 (2026) 745–754

Procedia Computer Science

www.elsevier.com/locate/procedia

The 6th International Conference on Machine Learning and Big Data Analytics for IoT Security and Privacy

Dynamic Adjustment Design of Intelligent Landscape Environmental Comfort from the Perspective of Human-Computer Interaction

Ruichao Zhao[a]*and Guoliang Li[b]

[a]*Department of Design and Art, Shandong Huayu University of Technology, Dezhou, 253034, Shandong, China*
[b]*Department of Information Engineering, Shandong Huayu University of Technology, Dezhou, 253034, Shandong, China*

Abstract

Current intelligent landscape environmental comfort adjustment is not sensitive to the physiological state of users and dynamic needs, and cannot achieve adjustment in time and personalization. This paper proposes a dynamic adjustment system of intelligent landscape environmental comfort designed on human-computer interaction. The system synchronously gathers parameters of the environment (temperature and humidity, lighting, noise) and user physiological behavior (heart rate variability, skin conductance, gait) by means of a heterogeneous sensor network. It uses the LSTM random forest hybrid model to implement multi-source data fusion and comfort prediction, and uses reinforcement learning algorithms to construct optimization control instructions of landscape equipment (shading, irrigation, lighting), which form a "perception analysis-decision execution" closed loop. The system has achieved 7.6 points average user comfort rating in 7 typical scenarios, 91.8% average personalized demand satisfaction rate, and 9.6 kWh average daily energy consumption, which has verified the effect of achieving accurate comfort improvement and energy efficiency synergy through dynamic interaction.

© 2026 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the SPIoT2025

Keywords: Human-Computer Interaction, Intelligent Landscape, Environmental Comfort, Dynamic Adjustment

1. Introduction

With the acceleration of urbanization and the increasing public demand for outdoor space quality, the comfort of landscape environment has become a key factor affecting urban livability and public health. Traditional landscape

* Corresponding author. Tel.: +0-000-000-0000 ; fax: +0-000-000-0000 .
E-mail address: ruichao3691@163.com

1877-0509 © 2026 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the SPIoT2025

environment design and management often rely on static and homogeneous physical index threshold control, or equipment scheduling based on fixed schedules. This type of method is difficult to adapt to the dynamic fluctuations of microclimate, the diversity of user activities, and the differences in individual comfort perception, resulting in delayed regulatory response, low energy efficiency, and inability to meet the growing demand for personalized experiences. In the context of the development of smart cities and IoT technology, how to deeply integrate real-time environmental perception, user status capture, and intelligent regulation of landscape facilities, and achieve a paradigm shift from "passive adaptation" to "active collaboration" in landscape environment, has become a cutting-edge challenge in the intersection of human-computer interaction and landscape design.

This article constructs a human centered intelligent landscape environment comfort dynamic adjustment system framework and conducts empirical verification: (1) This article proposes and implements a complete "multimodal perception data analysis intelligent decision-making IoT execution" closed-loop technology framework, which synchronously integrates user physiological and behavioral signals with environmental parameters as the core basis for quantifying and adjusting comfort decisions; (2) This article proposes a hybrid prediction model that combines LSTM temporal modeling and random forest regression to effectively capture the complex nonlinear and temporal dependencies of environment user interaction, thereby improving the accuracy of comfort assessment; (3) This article introduces a decision mechanism based on reinforcement learning, which balances comfort improvement and energy consumption costs through a custom reward function to achieve adaptive optimization of landscape regulation strategies. The system has demonstrated its comprehensive effectiveness in improving user subjective comfort, meeting personalized needs, and optimizing energy consumption through continuous experiments in diverse urban scenarios.

2. Related Work

The current research on landscape environment comfort has shifted from measuring single physical indicators to exploring multi-factor interaction mechanisms. Li et al. [1] used the central campus of Xuzhou Institute of Technology as an example to measure and analyze microclimate factors in different landscape spaces, and compared and analyzed the advantages and disadvantages of different landscape environment comfort levels. Xu and Zhou [2] explored the impact mechanism of landscape plant configuration on wind environment comfort, starting from landscape design, and proposed strategies to optimize the wind environment in residential areas, in order to enhance the comfort and ecological benefits of the living environment. Wang et al. [3] proposed a landscape design method to improve the comfort of fitness facilities in urban community parks by enriching the structure of plant communities, rationalizing vegetation, and reserving ventilation corridors. Under the background of smart city development, Çelik and Benliay [4] explored how technologies such as intelligent irrigation, sensor monitoring, and green infrastructure can be integrated with landscape design, and summarize existing practical models. The visual quality of urban space affects public experience and psychological feelings. Shedid and Hefnawy [5] analyzed the impact mechanism of landscape elements on visual elements in Egyptian urban public spaces, and verified it through case comparisons.

The intensification of urban heat island effect and the deterioration of microclimate affect energy consumption and public health, requiring cross scale landscape and urban design coordination to respond. Tavares and Fu [6] summarized microclimate regulation strategies from multiple dimensions, including urban form, blue-green infrastructure, and material albedo. Milani et al. [7] proposed that urban landscape design should enhance the diversity, accessibility, and participation of natural contact. Gaete Cruz et al. [8] proposed a collaborative design framework for urban landscapes. Li et al. [9] selected different landscape zones within Lanpu, Guangzhou to monitor their air temperature, relative humidity, negative oxygen ion concentration, air particulate matter concentration, and acoustic environmental factors, and analyze human comfort. Erdoğan et al. [10] proposed that carpet art can effectively enhance the cultural recognition and aesthetic value of landscape spaces.

Although some progress has been made in the analysis of elements, mechanism exploration, and technological integration of landscape comfort, existing research has failed to establish a closed loop between environmental perception, user feedback, and landscape efficacy. This article constructs a dynamic adjustment system with real-time monitoring, intelligent analysis, and interactive response as its core, promoting the paradigm shift of landscape environment from passive adaptation to active collaboration, and from homogeneous design to personalized

adaptation, in order to compensate for the shortcomings of current research in dynamics, interactivity, and systematicity.

3. Method

3.1 Dynamic Adjustment System Framework

The intelligent environment landscape comfort dynamic adjustment system proposed in this study is based on a closed-loop technical framework of "perception analysis decision execution" logic, as shown in Figure 1:

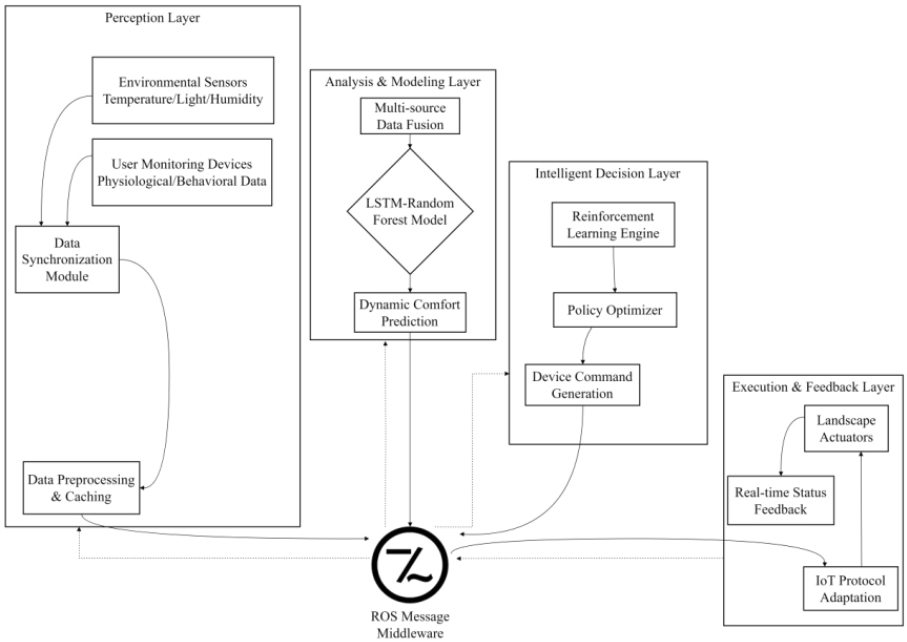


Fig 1. Intelligent Environment Landscape Comfort Dynamic Adjustment System

This framework focuses on human-computer interaction and aims to achieve a complete automated process from environmental and user state perception to precise control of landscape elements. The system starts from the multimodal perception layer and is responsible for synchronously collecting physical environment parameters and user physiological behavior data. These raw data are transmitted in real-time to the data analysis and modeling layer, where they are fused and parsed using LSTM random forest models to output dynamic user comfort prediction values. The predicted value serves as a key input to drive the reinforcement learning algorithm of the intelligent decision-making layer. Based on the current environmental state and user comfort goals, the algorithm calculates a set of optimal landscape equipment adjustment instructions. Finally, the execution and feedback layer regulates the various landscape execution mechanisms via IoT control protocols, such as servo motor driven sunshades, dimmable LED lamps, variable frequency water pump sprinkler systems to alter the microenvironment. At the same time, the environmental state and newly collected user data are fed back into the perception layer, forming a closed-loop feedback that enables the system to continuously optimize its adjustment strategy. The whole framework is integrated modularly accordingly to ROS for having low-latency and high-reliability data communication between layers.

3.2 Multimodal Perception Layer

The perception layer constructs a heterogeneous sensor network to comprehensively capture environmental status and human signals. The monitoring of environmental physical parameters adopts a distributed deployment strategy: high-precision digital temperature and humidity sensors (SHT35, sampling frequency 1Hz), illumination sensors (BH1750, range 0-65535 lux), and sound spectrum analysis modules (based on MAX9814 and FFT algorithm) are deployed in each experimental area, and data is transmitted through LoRa wireless network [11-12]. The sound spectrum analysis module performs a fast Fourier transform on the collected time-domain signal x(t), converting it into a frequency-domain representation to extract characteristic frequency components:

$$X(f)=\int_{-\infty}^{\infty} x(t)\, e^{-j2\pi ft}dt \tag{1}$$

X(f) refers to the complex spectrum of the signal at frequency f, x(t) refers to the original time-domain audio signal that varies over time t, and $e^{-j2\pi ft}$ is a complex exponential form kernel function. This transformation decomposes time-domain audio signals into amplitude and phase information of different frequencies, which is used for subsequent identification of environmental noise levels or specific sound events.

User side data collection relies on wearable devices (customized wristbands, built-in PPG photoelectric heart rate sensor and GSR skin conductance sensor, sampling rate of 64Hz) and smartphones with built-in three-axis accelerometers. The wristband uploads the time-domain indicators of heart rate variability (HRV) (RMSSD) and skin conductance level (SCL) raw values at 1-second intervals via Bluetooth 5.0; Mobile accelerometer data is used to infer user activity states (stationary, walking, running) through gait analysis algorithms (based on time window peak detection). After all data streams are stamped with a unified time stamp, they are real-time converged to the edge computing gateway (using NVIDIA Jetson Xavier NX) through the MQTT protocol for Kalman filtering to smooth the temperature series and eliminate outliers. The Kalman filter performs optimal estimation on temperature observation sequences with noise through two steps: state prediction and update. The state update equation is:

$$\hat{x}_{k\mid k}=\hat{x}_{k\mid k-1}+K_k(z_k-H\hat{x}_{k\mid k-1}) \tag{2}$$

$\hat{x}_{k\mid k}$ is the estimated state value at time k, K_k is the Kalman gain, z_k is the actual observation value, and H is the observation matrix. This process effectively filters out random noise in environmental temperature data, improving data reliability.

3.3 Data Analysis and Modeling Layer

This layer converts multi-source heterogeneous data into quantifiable indicators of user comfort that can guide decision-making. Firstly, the data fusion module aligns the environmental parameter sequence with the user's physiological signal sequence using a 5-second time window, and concatenates them into a feature vector. The features include the current window's average temperature and humidity, average light intensity, noise energy in the 0-500Hz frequency band, user HRV average, SCL average, and gait state encoding. Subsequently, the feature vectors are input into an improved LSTM random forest hybrid model. The specific operation logic of the model is as follows: the original temporal features (temperature, humidity, lighting, noise, HRV, SCL) are first input into a two-layer LSTM network (64 hidden units per layer, dropout rate 0.2) to capture the dynamic temporal dependencies between the environment and physiological signals. The output of the last time step of the LSTM is used as the advanced temporal feature [13]. The core computation of the LSTM unit at each time step t involves updating the cell state, and its key gating mechanism is controlled by the forget gate f_t, which determines how much information is retained from the previous cell state:

$$f_t=\sigma(W_f\cdot[h_{t-1},r_t]+b_f) \tag{3}$$

σ refers to the sigmoid activation function, W_f and b_f are the weight matrix and bias vector of the forget gate, h_{t-1} is the hidden state of the previous time step, and r_t is the input feature vector of the current time step.

This feature is then concatenated with features representing the user's static attributes (such as age group encoding) and current behavior (gait state) to form the final feature set. This set is fed into a random forest model (consisting of 200 decision trees with a maximum depth of 15, using Gini impurity as the splitting criterion). When constructing each decision tree, the Gini impurity used to evaluate the quality of node splitting is:

$$I_G(p)=1-\sum_{i=1}^{c} p_i^2 \tag{4}$$

c is the number of categories for the target variable (comfort score), and p_i is the proportion of samples belonging to the i-th category in the node.

The goal of random forest training is to regress and predict the real-time comfort level of users (output a rating of 0-10), and its final output is the average of the predicted values of all decision trees:

$$\hat{y}=\frac{1}{M}\sum_{m=1}^{M} T_m(x) \tag{5}$$

$\hat{y}$ is the final comfort prediction value of the random forest for the input feature vector x, M is the total number of decision trees (here 200), and $T_m(x)$ refers to the predicted output of the mth decision tree for feature x.

Random forests are great for non-linear interactions between high dimensional features and can reveal mappings between patterns of user behavior, and both historical preferences, and complex combinations of environment. The model is trained on data annotated by the user in the last 30 days (collected via an app of user feedback) and updated daily.

3.4 Intelligent Decision-making Layer

The decision-making layer uses a Q-learning-based reinforcement learning algorithm to generate specific control commands for landscape equipment. This algorithm employs a Markov decision process model for environmental regulation problems. The state space S consists of the following elements: normalized current environmental physical parameters (temperature, humidity, light intensity, noise), predicted user comfort scores, and temporal context. The action space A is defined as a set of discrete combinations of landscape equipment controls, including but not limited to: {shading angle: 0%, 50%, 100%; water spray intensity: off, low, high; main lighting color temperature: 3000K, 4000K, 5000K} [14-15]. The reward function R is calculated considering improvements in user comfort, energy costs, and the smoothness of equipment actions.

$$R_t=\alpha\cdot(C_t-C_{t-1})-\beta\cdot E_t-\gamma\cdot P_t \tag{6}$$

R_t refers to the immediate reward obtained after performing an action at time t, C_t and C_{t-1} respectively refer to the predicted user comfort scores at the current and previous time points, E_t refers to the estimated energy consumption for performing the current action, P_t refers to the penalty term for the amplitude of the device action, and α, β, and γ are the weight coefficients for each sub item.

The search process focuses on varying weight combinations and executes the algorithm in a simulated environment for a specified period of time and analyses its average cumulative reward and strategy stability. Representative search examples can be seen in Table 1:

Table 1. Reward Function Weight Grid Search Results

Comfort Improvement Weight	Energy Cost Weight	Action Smoothness Weight	Simulated Average Cumulative Reward
0.9	0.1	0.0	85.2
0.8	0.1	0.1	92.7
0.7	0.2	0.1	96.3
0.6	0.3	0.1	88.5
0.5	0.4	0.1	79.1
0.7	0.1	0.2	90.4

Grid search results show that the algorithm achieves the highest average cumulative reward (96.3) in the simulated environment when the weight combination is α=0.7, β=0.2, γ=0.1. This means that this weight

configuration optimally balances improving comfort, suppressing excessive energy consumption, and reducing frequent device movement. Ultimately, the weights of the reward function are set to α=0.7, β=0.2, γ=0.1.

The algorithm maintains a Q-table and selects the action that maximizes the expected cumulative reward at each decision time (per 10-second interval) according to an ε-greedy policy (initial value 0.3, decaying over time). After executing the action, the system acquires another state and calculates the reward to update the Q-value (learning rate 0.1, discount factor 0.9). The algorithm is pre-trained in a simulated environment and then run online to continuously optimize the policy. For the ε-greedy policy, the probability of selecting a certain action is determined by the exploration rate:

$$\pi(a \mid s)=\begin{cases}1-\epsilon+\frac{\epsilon}{|A|} & \text{if } a=\arg\max_{a'} Q(s,a') \\ \frac{\epsilon}{|A|} & \text{otherwise}\end{cases} \tag{7}$$

$\pi(a \mid s)$ refers to the probability of selecting action a in state s; ϵ is the exploration rate, initially 0.3 and decreasing over time, $|A|$ is the size of the action space; $Q(s,a')$ is the Q function, which refers to the estimated expected cumulative return of selecting action a' in state s. The strategy ensures that the algorithm selects the current optimal action with a high probability, while preserving a certain degree of randomness to explore better strategies.

3.5 Execution and Feedback Layer

This layer is the interface between the system and physical landscape devices which allow IoT execution and feedback decision instructions. The core is an IoT control platform, which is based on Node RED. The platform uses the MQTT protocol to communicate with a variety of landscape devices controllers. For example: Servo motor controller of sunshade gets 0-100 percent of open and close angle command. Intelligent lighting driver gets the parameters of brightness and light color in the form of JSON. Solenoid valve and frequency variable of sprinkler irrigation system get the setting parameters of sprinkler irrigation flow and irrigation duration. After receiving action instructions from the decision-making layer, the platform decomposes the instructions into specific device instructions and synchronously issues them. In order to make sure the execution reliability, the platform has introduce instruction confirmation and status query mechanism, and each device need to send a status confirmation frame after the action executed. The feedback loop is established through two levels: one is that the device's own state feedback is directly transmitted back to the decision-making layer for correcting the device model; The second is the continuous monitoring and adjustment of physical parameter changes by the environmental sensor network. These new parameters, along with user physiological data, are used as inputs for the next perception cycle to evaluate the actual effect of the adjustment action.

$$e(t)=y_{set}(t)-y_{fb}(t) \tag{8}$$

e(t) is the tracking error at time t, $y_{set}(t)$ is the set value of the device parameters, and $y_{fb}(t)$ is the actual measurement value fed back by the sensor. This error is used for continuous calibration of device model parameters, forming a complete closed loop from "decision" to "environmental change" and then to "new perception" [16]. The end-to-end delay of the entire instruction from issuance to status confirmation is controlled within the design target of 800 milliseconds.

4. Results and Discussion

4.1 Experimental Design

Three typical urban outdoor spaces were selected as test scenarios for the experiment: city square, community park, commercial district, campus tree lined avenue, waterfront ecological corridor, hospital rehabilitation garden, and commercial office atrium. Within the radius of 50 meters of each scene, three sensor nodes (including temperature and humidity, light and noise sensors) and two edge computing gateways are deployed. This article recruited 90 participants (30 per scenario, half male and half female, aged 20-65), wearing customized wristbands

and installing a data collection app. The experiment can be conducted continuously for 30 days (covering sunny, rainy, cloudy and other weather conditions), with the system running from 8:00 to 20:00 daily. The evaluation indicators are quantified through sensor data, device energy consumption measurement sockets, and a 0-10 point comfort rating submitted by participants every 30 minutes through the APP. Comparative experiments were conducted synchronously: Baseline 1 used a fixed schedule (sunshade unfolded at 8:00/retracted at 18:00, sprinkler irrigation for 5 minutes each at 12:00 and 18:00 daily) and threshold control (sprinkler irrigation started when temperature>28 ℃); Baseline 2 uses the same sensor network but only triggers device adjustments based on environmental parameters (temperature, humidity, light, noise) through preset rules, without accessing user physiological data. All systems run under the same hardware and experimental cycle.

4.2 Experimental Results

The comfort score is obtained through subjective feedback (0-10 points) from participants' daily periodic (every 30 minutes) APP. Average the daily ratings of each participant, and then average the daily ratings of all participants to obtain the overall 30 day average comfort rating for each system in each scenario (Figure 2):

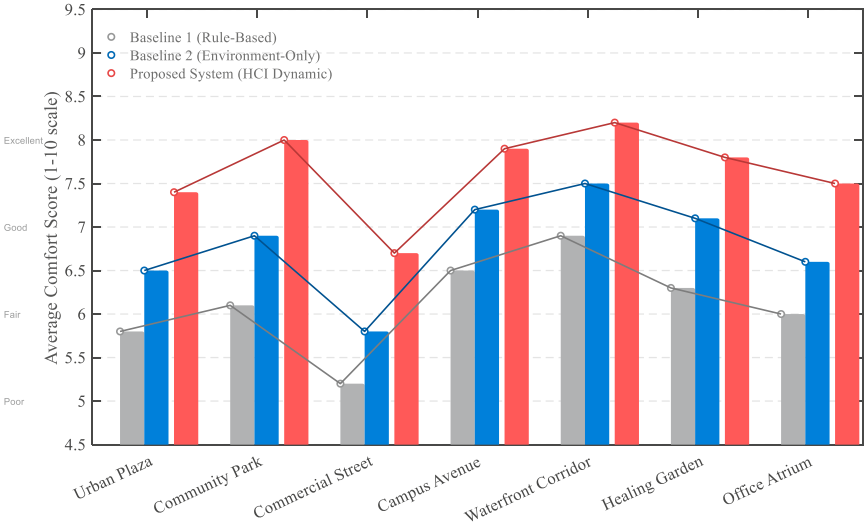


Fig 2. User Comfort Rating

The system in this article achieved the highest user comfort rating in all seven testing scenarios, with an overall average score of 7.6 points significantly higher than the baseline 1 system (6.1 points) and baseline 2 system (6.8 points). Compared with the fixed baseline 1 system, the comfort improvement ranges from 18.8% to 31.1%, with the community park scene showing the greatest improvement, which is related to the diverse types of user activities (sitting, walking, fitness), highlighting the advantage of personalized adaptation through behavior recognition in the system. The waterfront ecological corridor has relatively small improvement due to its favorable microclimate conditions and high basic comfort. Compared to the baseline 2 system that relies solely on environmental parameters, the advantages of this system are equally evident, demonstrating that integrating user physiological and behavioral signals can more accurately capture and respond to their comfort needs, compensating for the lag and one sidedness of simple environmental feedback.

Energy consumption data is collected through smart sockets connected to various landscape equipment (shading motors, water pumps, lighting fixtures), and the total energy consumption of each system during the 30 day experimental period is summarized and converted into daily average energy consumption (kWh/day) for fair comparison. The specific energy consumption data is shown in Figure 3:

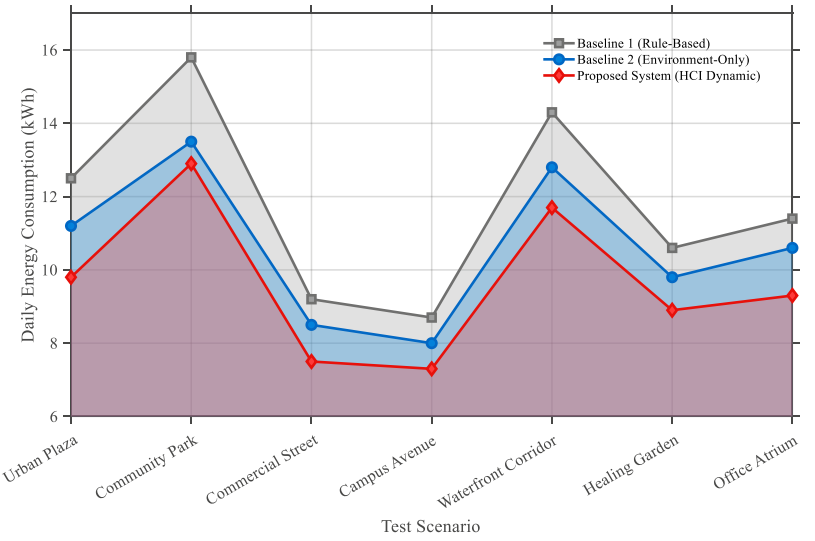


Fig 3. Daily Energy Consumption

The energy consumption data shows that the system described in this paper improves comfort while achieving certain energy-saving effects. However, its energy consumption (9.6 kWh/day) is 18.6% lower than the baseline system 1 (11.8 kWh/day). This is mainly attributed to the introduction of an energy cost penalty term (weight β=0.2) into the reward function (Equation 6) of the reinforcement learning decision model. This penalty term aims to guide the system to select a more energy-efficient combination of equipment actions (while also considering comfort) while meeting comfort goals. For example, during periods of strong sunlight, such as in the afternoon, the system can prioritize adjusting shading measures rather than excessively activating sprinkler cooling. The baseline 2 system, due to its responsive adjustment, avoids the ineffective timed operation of baseline 1 and reduces energy consumption by 10.6%. However, it lacks perception of the user's true state and sometimes generates redundant adjustments, resulting in lower energy-saving effects compared to the system in this paper.

4.3 Personalized Requirement Satisfaction Rate

The personalized demand satisfaction rate is calculated by recording the system's automatic adjustment actions and the user's manual intervention behavior within the next 15 minutes. Figure 4 shows the average satisfaction rate of different user groups (by activity type) in various scenarios, and the overall average satisfaction rate of the system reaches 91.8%.

There are certain differences between different scenarios, and the fundamental reason lies in the differences in scene functions, user group characteristics, and environmental interference levels. In hospital rehabilitation gardens (96.3%) and community parks (94.8%), the satisfaction rate is high. This is because the user activities in these scenarios have clear and relatively stable purposes, and the environment is relatively quiet and controllable. The multimodal perception layer can accurately capture the user's physiological signals, and the intelligent decision-making layer can generate highly matched adjustment strategies based on this, achieving precise personalized adaptation. Based on the analysis of scenario differences in the satisfaction rate of personalized needs, the dynamic adjustment system constructed in this study, with human-computer interaction as the core, can fully leverage its technological advantages and achieve extremely high satisfaction rates of personalized needs in an "inward oriented" landscape space with relatively stable environment and clear user behavior patterns. This confirms the effectiveness of the system core framework in accurately understanding and adapting individual comfort needs

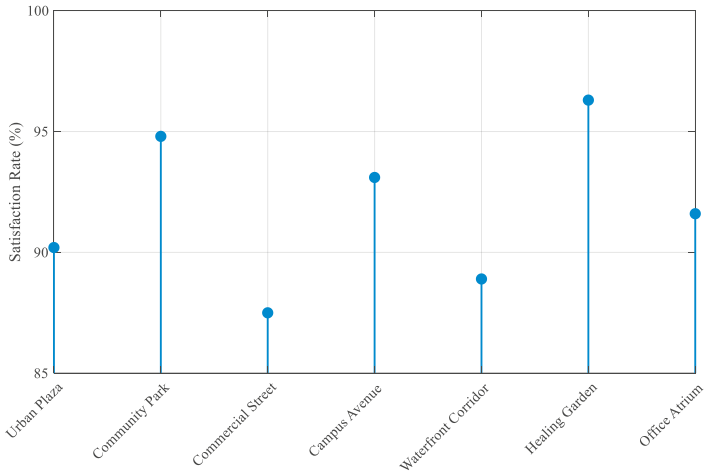


Fig 4. Personalized demand satisfaction rate under different types of user activities

5. Conclusion

This article constructs an intelligent landscape environment comfort dynamic adjustment system with human-computer interaction as the core. By integrating multimodal environment and user data, and using LSTM random forest model and reinforcement learning algorithm, the landscape environment is transformed from static homogeneous control to dynamic personalized adaptation. However, the robustness of the system under extreme weather conditions, the generalization ability of long-term user behavior patterns, and the cost-effectiveness of larger scale deployment still need to be further verified. Future work can explore the deployment of lightweight models, multi-agent collaborative regulation mechanisms, and further integrate social and psychological dimension parameters to promote the development of intelligent landscapes towards greater inclusiveness, resilience, and empathy.

References

[1] Li Kewei, Chi Xiaoli, An Qi. Research on Microclimate and Environmental Comfort of Different Campus Landscapes - A Case Study of Xuzhou University of Technology [J]. Modern Horticulture, 2025, 48(13): 63-65

[2] Xu Yao, Zhou Xiaohui. Landscape Design Strategy for Shenzhen Affordable Housing Based on Wind Environment Comfort - A Case Study of Shenkang Village [J]. Footwear Technology and Design, 2025, 5(20): 133-135

[3] Wang Qing, Li Meng, Li Xiangyi. Research on Landscape Design of Community Park Fitness Facilities Based on Microclimate Human Comfort [J]. Chinese Landscape Architecture,2021,37(8):68-73

[4] Çelik F T, Benliay A. SMART CITY APPLICATIONS AND THE ROLE OF LANDSCAPE DESIGN IN MASS HOUSING: THE FUTURE OF URBAN LIFE[J]. JOURNAL OF LANDSCAPE RESEARCH AND PRACTICES (JOLARP), 2025, 7(1): 13-20.

[5] Shedid M Y, Hefnawy N H. Influence of landscape elements on visual design elements in order to enhance the visual quality of urban spaces[J]. Journal of Engineering Research, 2022, 6(4): 127-134.

[6] Tavares S, Fu J. Landscape and urban design for improved urban microclimate[J]. Landscape Review, 2025, 21(2): 38-45.

[7] Milani A M, Golmohammadi M, Hoveyzavi A, et al. Linking Urban Landscape Design, Health, and Sustainability: Insights from Nature-Based and Environmental Psychology Approaches[J]. Journal of Salutogenic Architecture, 2025, 4(1): 248-256.

[8] Gaete Cruz M, Ersoy A, Czischke D, et al. Towards a framework for urban landscape co-design: Linking the participation ladder and the design cycle[J]. CoDesign, 2023, 19(3): 233-252.

[9] Li Bingmin, Ma Mianying, Xie Weiwen, Dong Chenlu, Tan Guangwen. Study on environmental factors and human comfort in Guangzhou Orchid Garden [J]. Guangdong Landscape Architecture,2024,46(4):60-66

[10] Erdoğan R, Demirhan D N, Olgun R. Carpet Art As A Source of Inspiration in Urban Landscape Design[J]. COMU Journal of Agriculture Faculty, 2022, 10(2): 284-293.

[11] Jiang Haobo, Song Mengkun, Xiao Yang. Research on the Evaluation of Visual Landscape Comfort in Waterfront Spaces—Taking the "One River and One Canal" in Shanghai as an Example [J]. Landscape Architecture,2022,29(10):122-129

[12] Wang Qing, Sun Haodong, Xie Xiaohuan. Landscape optimization of park recreational spaces based on microclimate comfort [J]. China Urban Forestry,2022,20(5):10-15

[13] Luka N. Walking beyond the city? On the importance of recreational mobilities for landscape planning, urban design, and public policy[J]. Mobilities, 2023, 18(5): 789-804.

[14] ÇAKIR M. Şehit Ömer Halisdemir Square Urban Design and Landscape Design: Şehit Ömer Halisdemir Meydanı Kentsel Tasarım ve Peyzaj Tasarımı[J]. Journal of Protected Areas Research, 2023, 2(2): 72-84.

[15] Vasiliev D, Stevens R, Bornmalm L, et al. THE ROLE OF LANDSCAPE DESIGN AND URBAN NATURE-BASED SOLUTIONS IN ADDRESSING GLOBAL CHALLENGES[J]. International Multidisciplinary Scientific GeoConference: SGEM, 2023, 23(6.2): 721-728.

[16] Quistberg D A, Hessel P, Rodriguez D A, et al. Urban landscape and street-design factors associated with road-traffic mortality in Latin America between 2010 and 2016 (SALURBAL): an ecological study[J]. The Lancet Planetary Health, 2022, 6(2): e122-e131.